

ROLL NUMBER : CT-25169

QUESTION : 1

```
#include <iostream>
#include <cstring>
using namespace std;

class Employee {
private:
    char* EmployeeName;
    const int EmployeeId;

public:
    Employee(const char* name, int id) : EmployeeId(id) {
        EmployeeName = new char[strlen(name) + 1];
        strcpy(EmployeeName, name);
    }

    // Destructor
    ~Employee() {
        delete[] EmployeeName;
    }

    // Getter for EmployeeName
    const char* getEmployeeName() const {
        return EmployeeName;
    }

    // Getter for EmployeeId
    int getEmployeeId() const {
        return EmployeeId;
    }

    // Setter for EmployeeName
    void setEmployeeName(const char* name) {
        delete[] EmployeeName;
        EmployeeName = new char[strlen(name) + 1];
        strcpy(EmployeeName, name);
    }
};

int main() {

    Employee Employee1("Ali", 101);
    Employee Employee2("Sara", 102);
    Employee Employee3("Ahmed", 103);

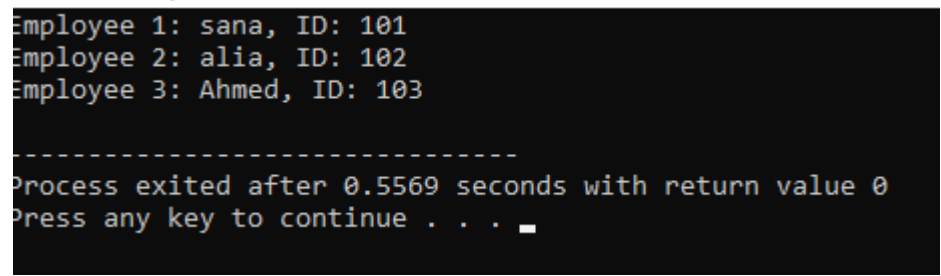
    // Changing names
    Employee1.setEmployeeName("sana");
    Employee2.setEmployeeName("alia");
```

```
cout << "Employee 1: " << Employee1.getEmployeeName()
    << ", ID: " << Employee1.getEmployeeId() << endl;

cout << "Employee 2: " << Employee2.getEmployeeName()
    << ", ID: " << Employee2.getEmployeeId() << endl;

cout << "Employee 3: " << Employee3.getEmployeeName()
    << ", ID: " << Employee3.getEmployeeId() << endl;

return 0;
}
```



```
Employee 1: sana, ID: 101
Employee 2: alia, ID: 102
Employee 3: Ahmed, ID: 103

-----
Process exited after 0.5569 seconds with return value 0
Press any key to continue . . .
```

QUESTION : 2

```
#include <iostream>
using namespace std;

class DynamicArray {
private:
    int* arr;
    int capacity;
    int currentSize;

public:
    // Parameterized Constructor
    DynamicArray(int size) {
        capacity = size;
        currentSize = 0;
        arr = new int[capacity];

        for (int i = 0; i < capacity; i++) {
            arr[i] = 0; // Initialize all values with 0
        }
    }

    void push(int value) {
        if (currentSize < capacity) {
            arr[currentSize] = value;
            currentSize++;
        } else {
            cout << "Array is Full!" << endl;
        }
    }
}
```

```
// size() function
int size() {
    return currentSize;
}

// Display function
void display() {
    for (int i = 0; i < currentSize; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

// Destructor
~DynamicArray() {
    delete[] arr;
}
};

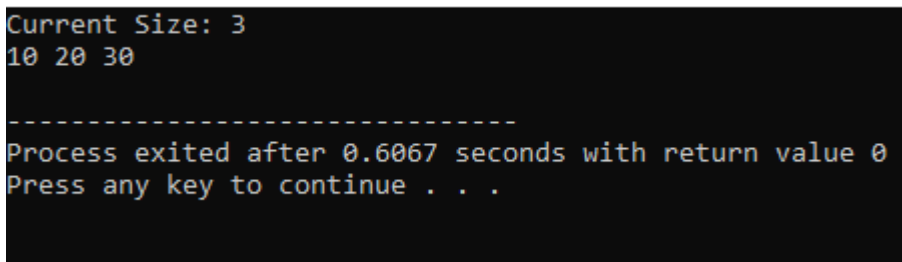
int main() {
    DynamicArray obj(5);

    obj.push(10);
    obj.push(20);
    obj.push(30);

    cout << "Current Size: " << obj.size() << endl;
    obj.display();

    return 0;
}
```

 C:\Users\Hp\Documents\lab 4.exe



```
Current Size: 3
10 20 30

-----
Process exited after 0.6067 seconds with return value 0
Press any key to continue . . .
```

QUESTION : 3

```
#include <iostream>
using namespace std;

class Account {
private:
    int account_no;
    double account_bal;
    int security_code;
```

```
static int count; // Static variable

public:
    // Function to initialize
    void initialize(int no, double bal, int code) {
        account_no = no;
        account_bal = bal;
        security_code = code;
        count++; // Increase object count
    }

    // Function to display data
    void display() {
        cout << "Account No: " << account_no << endl;
        cout << "Balance: " << account_bal << endl;
        cout << "Security Code: " << security_code << endl;
        cout << "Total Objects: " << count << endl;
        cout << "--" << endl;
    }
};

// Initialize static variable
int Account::count = 0;

int main() {

    Account a1, a2;

    a1.initialize(101, 5000.50, 1234);
    a2.initialize(102, 8000.75, 5678);

    a1.display();
    a2.display();

    return 0;
}
```

```
Account No: 101
Balance: 5000.5
Security Code: 1234
Total Objects: 2
--
Account No: 102
Balance: 8000.75
Security Code: 5678
Total Objects: 2
--
-----
Process exited after 0.5411 seconds with return value 0
Press any key to continue . . .
```

QUESTION : 4

```
#include <iostream>
using namespace std;

class Student {
private:
    string name;
    const int rollNo; // constant data member

public:
    // Constructor using member initializer list
    Student(string n, int r) : rollNo(r) {
        name = n;
    }

    // Setter for name (can change)
    void setName(string n) {
        name = n;
    }

    // Getter (const function)
    string getName() const {
        return name;
    }

    // Getter for roll number (const function)
    int getRollNo() const {
        return rollNo;
    }
};

int main() {
```

```
    Student s1("Ali", 101);
```

```
    s1.setName("Ahmed"); // Name can change
```

```
    cout << "Name: " << s1.getName() << endl;
    cout << "Roll No: " << s1.getRollNo() << endl;
```

```
    return 0;
}
```

```
Name: Ahmed
Roll No: 101
```

```
-----
Process exited after 0.4986 seconds with return value 0
Press any key to continue . . .
```

QUESTION : 5

```
#include <iostream>
using namespace std;

// Interface
class Hotel {
public:
    virtual void calculateRent() = 0;
    virtual void display() const = 0; // const function
};

// Implementation class
class HotelMercato : public Hotel {
private:
    string customerName;
    int days;
    double rent;
    const double rate; // Fixed rate (cannot change)

public:
    // Constructor using member initializer list
    HotelMercato(string name, int d)
        : customerName(name), days(d), rate(1000.85), rent(0) {}

    // Rent calculation
    void calculateRent() override {
        if (days > 7) {
            rent = (days - 1) * rate; // Discount: subtract 1 day
        } else {
            rent = days * rate;
        }
    }

    // Display function (cannot modify data)
    void display() const override {
        cout << "Customer Name: " << customerName << endl;
        cout << "Days Stayed: " << days << endl;
        cout << "Total Rent: " << rent << endl;
    }
};

int main() {

    HotelMercato customer1("Ali", 5);
    HotelMercato customer2("Sara", 10);

    customer1.calculateRent();
    customer2.calculateRent();

    customer1.display();
    cout << "-----" << endl;
    customer2.display();

    return 0;
}
```

```
Customer Name: Ali
Days Stayed: 5
Total Rent: 5004.25
-----
Customer Name: Sara
Days Stayed: 10
Total Rent: 9007.65
-----
Process exited after 0.6192 seconds with return value 0
Press any key to continue . . .
```